8 **(a)** The curve $y = \dfrac{1}{(1+x^2)^2}$ is shown in Fig. 8.

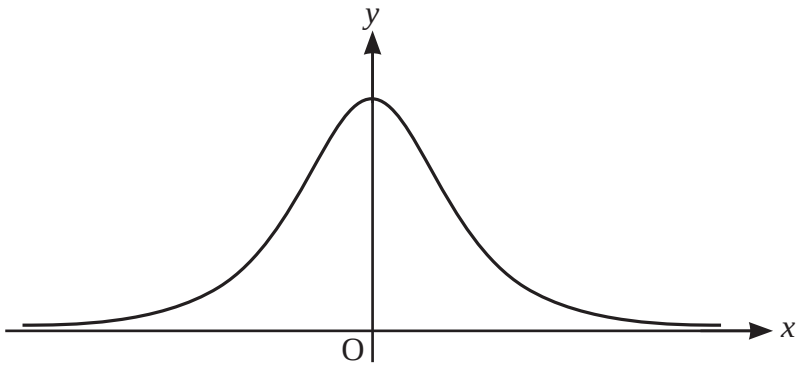

Fig. 8

(i) Show that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{20x^2-4}{(1+x^2)^4}$. **[5]**

(ii) **In this question you must show detailed reasoning.**

Find the set of values of x for which the curve is concave downwards. **[3]**

(b) Use the substitution $x = \tan\theta$ to find the exact value of $\displaystyle\int_{-1}^{1} \frac{1}{(1+x^2)^2}\,\mathrm{d}x$. **[8]**